```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from google.colab import files
```

```
uploaded = files.upload()
```

Choose Files 2 files

MuRIL_Evaluation_Results.csv(text/csv) - 102 bytes, last modified: 11/6/2026 - 100% done
Model_Evaluation_Results.csv(text/csv) - 156 bytes, last modified: 11/6/2026 - 100% done
Saving MuRIL_Evaluation_Results.csv to MuRIL_Evaluation_Results.csv
Saving Model_Evaluation_Results.csv to Model_Evaluation_Results.csv

```
ml_results = pd.read_csv("Model_Evaluation_Results.csv")
muril_results = pd.read_csv("MuRIL_Evaluation_Results.csv")

print("Step 8 results:")
display(ml_results)

print("\nMuRIL results:")
display(muril_results)
```

Step 8 results:

	Model	Accuracy	Precision	Recall	F1_Score
0	Logistic Regression	100.0	100.0	100.0	100.0
1	Naive Bayes	100.0	100.0	100.0	100.0
2	Linear SVM	100.0	100.0	100.0	100.0

MuRIL results:

	Model	Accuracy	Precision_Misinformation	Recall_Misinformation	F1_Misinfo
0	MuRIL	1.0	1.0	1.0	

Next steps: Generate code with `ml_results` New interactive sheet

```
print("ML columns:", ml_results.columns.tolist())
print("MuRIL columns:", muril_results.columns.tolist())
```

```
ML columns: ['Model', 'Accuracy', 'Precision', 'Recall', 'F1_Score']
MuRIL columns: ['Model', 'Accuracy', 'Precision_Misinformation', 'Recall_Misinfo
```

```
# Standardize Step 8 column names
ml_results.columns = ml_results.column

# Standardize MuRIL column names
muril_results.columns = muril_results.

print("Traditional ML:")
display(ml_results.head())

print("MuRIL:")
display(muril_results.head())
```

Traditional ML:

	Model	Accuracy	Precision	Recall	F1_Score
0	Logistic Regression	100.0	100.0	100.0	100.0
1	Naive Bayes	100.0	100.0	100.0	100.0
2	Linear SVM	100.0	100.0	100.0	100.0

MuRIL:

	Model	Accuracy	Precision_Misinformation	Recall_Misinformation	F1_Misinfo
0	MuRIL	1.0	1.0	1.0	

```
print("STEP 8 COLUMNS:")
print(ml_results.columns.tolist())

print("\nSTEP 8 DATA:")
display(ml_results)

print("\nSTEP 9 COLUMNS:")
print(muril_results.columns.tolist())

display(muril_results)
```

```
STEP 8 COLUMNS:
['Model', 'Accuracy', 'Precision', 'Recall', 'F1_Score']

STEP 8 DATA:
```

	Model	Accuracy	Precision	Recall	F1_Score
0	Logistic Regression	100.0	100.0	100.0	100.0
1	Naive Bayes	100.0	100.0	100.0	100.0
2	Linear SVM	100.0	100.0	100.0	100.0

```
STEP 9 COLUMNS:
['Model', 'Accuracy', 'Precision_Misinformation', 'Recall_Misinformation', 'F1_M
```

	Model	Accuracy	Precision_Misinformation	Recall_Misinformation	F1_Misinfo
0	MuRIL	1.0	1.0	1.0	

Next steps: Generate code with `ml_results` New interactive sheet

```
# Step 8: Traditional ML models
ml_comparison = ml_results[
    ["Model", "Accuracy", "Precision", "Recall", "F1_Score"]
].copy()

# Rename F1_Score to F1
ml_comparison.rename(
    columns={"F1_Score": "F1"},
    inplace=True
)

# Step 9: MuRIL Transformer
muril_comparison = pd.DataFrame({
    "Model": ["MuRIL"],
    "Accuracy": [muril_results["Accuracy"].iloc[0] * 100],
    "Precision": [muril_results["Precision_Misinformation"].iloc[0] * 100],
    "Recall": [muril_results["Recall_Misinformation"].iloc[0] * 100],
    "F1": [muril_results["F1_Misinformation"].iloc[0] * 100]
})

# Combine all models
final_comparison = pd.concat(
    [ml_comparison, muril_comparison],
    ignore_index=True
)
```

```
# Display comparison
display(final_comparison)
```

	Model	Accuracy	Precision	Recall	F1
0	Logistic Regression	100.0	100.0	100.0	100.0
1	Naive Bayes	100.0	100.0	100.0	100.0
2	Linear SVM	100.0	100.0	100.0	100.0
3	MuRIL	100.0	100.0	100.0	100.0

Next steps: Generate code with `final_comparison` New interactive sheet

```
final_comparison[
    ["Accuracy", "Precision", "Recall"
] = final_comparison[
    ["Accuracy", "Precision", "Recall"
].astype(float)

final_comparison = final_comparison.ro

display(final_comparison)
```

	Model	Accuracy	Precision	Recall	F1
0	Logistic Regression	100.0	100.0	100.0	100.0
1	Naive Bayes	100.0	100.0	100.0	100.0
2	Linear SVM	100.0	100.0	100.0	100.0
3	MuRIL	100.0	100.0	100.0	100.0

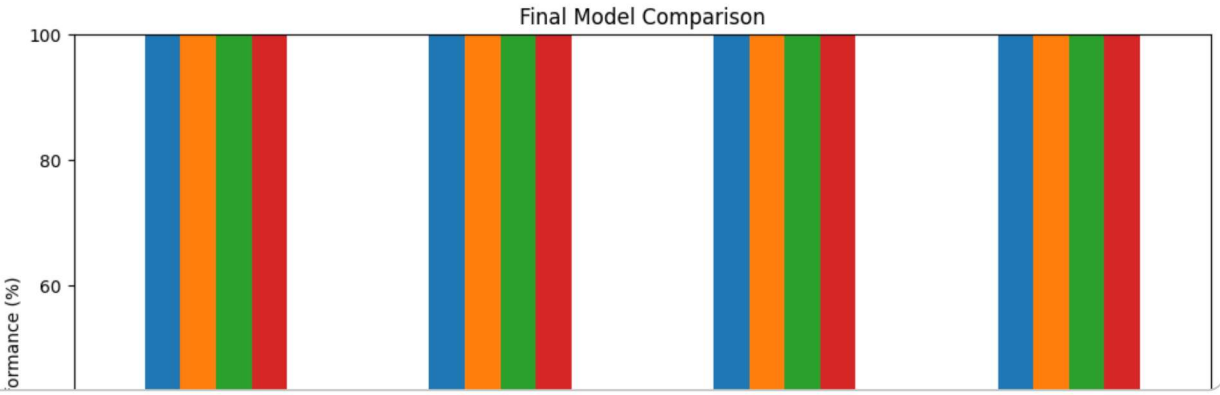
Next steps: Generate code with `final_comparison` New interactive sheet

```
final_comparison.set_index("Model")[
    ["Accuracy", "Precision", "Recall", "F1"]
].plot(kind="bar", figsize=(10, 6))

plt.title("Final Model Comparison")
plt.xlabel("Machine Learning Models")
```

```
plt.ylabel("Performance (%)")
plt.ylim(0, 100)
plt.xticks(rotation=0)
plt.legend(title="Metrics")
plt.tight_layout()

plt.savefig("Final_Model_Comparison.png", dpi=300)
plt.show()
```




```
final_comparison = final_comparison.sort_values(
    by="F1",
    ascending=False
)

display(final_comparison)

best_model_name = final_comparison.iloc[0]["Model"]

print("Selected Final Model:", best_model_name)
print("Selection criterion: Highest MISINFORMATION F1-score")
```

	Model	Accuracy	Precision	Recall	F1
0	Logistic Regression	100.0	100.0	100.0	100.0
1	Naive Bayes	100.0	100.0	100.0	100.0
2	Linear SVM	100.0	100.0	100.0	100.0
3	MuRIL	100.0	100.0	100.0	100.0

```
Selected Final Model: Logistic Regression
Selection criterion: Highest MISINFORMATION F1-score
```

Next steps: Generate code with `final_comparison` New interactive sheet

```
final_comparison.to_csv(
```

```
    "Final_Model_Comparison.csv",
    index=False
)

selection = pd.DataFrame({
    "Selected_Model": [best_model_name
    "Selection_Criterion": ["Highest M
})

selection.to_csv(
    "Final_Model_Selection.csv",
    index=False
)

print("Final model selection saved")
```

```
Final model selection saved
```

```
wnload("Final_Model_Comparison.csv")
wnload("Final_Model_Selection.csv")
wnload("Final_Model_Comparison.png")
```